

№4

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1. NS-2;
2. TCP (Xgraph GNUPlot);
3. ;
4. .

:

- N TCP- , N TCP- , R1 R2
(N — 20);
- TCP- 100
/ 20 DropTail;
- TCP- 100
/ 20 DropTail;

- 15 RED, 300 ; — 20 / (R1-R2) / (R2-R1)
 15 / 20 DropTail;
- FTP TCPReno;
- RED: $q_{min} = 75, q_{max} = 150, q_w = 0,002, p_{max} = 0.1$;
- TCP- 32; 500 ; — 20
 .

```
#           Simulator
set ns [new Simulator]
#           out.nam           nam
set nf [open out.nam w]
#           nf
$ns namtrace-all $nf
#           out.tr
#
set f [open out.tr w]
#           f
$ns trace-all $f
```

```
Agent/TCP set window_ 32
Agent/TCP set pktSize_ 500
```



```
#      finish
proc finish {} {
    global tchan_
    #      AWK:
    set awkCode {
    {
        if ($1 == "Q" && NF>2) {
            print $2, $3 >> "temp.q";
            set end $2
        }

        else if ($1 == "a" && NF>2)
            print $2, $3 >> "temp.a";
        }
    }
}
```

```
exec rm -f temp.q temp.a
```

```
exec touch temp.a temp.q
```

```
exec awk $awkCode all.q
```

```
#      xgraph          TCP      :
```

```
exec xgraph -fg pink -bg purple -bb -tk -x time -t "TCPRenoCWND" WindowVsT
```

```
exec xgraph -fg pink -bg purple -bb -tk -x time -t "TCPRenoCWND" WindowVsT
```

```
exec xgraph -bb -tk -x time -y queue temp.q &
```

```
exec xgraph -bb -tk -x time -y queue temp.a &
```

```
exec nam out.nam &
```

```
exit 0
```

```
}
```

```
#                                TCP:
proc plotWindow {tcpSource file} {
    global ns
    set time 0.01
    set now [$ns now]
    set cwnd [$tcpSource set cwnd_]
    puts $file "$now $cwnd"
    $ns at [expr $now+$time] "plotWindow $tcpSource $file"
}
```

```
set r1 [$ns node]
set r2 [$ns node]
$ns simplex-link $r1 $r2 20Mb 15ms RED
$ns simplex-link $r2 $r1 15Mb 20ms DropTail
$ns queue-limit $r1 $r2 300
```

```
set N 30
for {set i 0} {$i < $N} {incr i} {
    set n1($i) [$ns node]
    $ns duplex-link $n1($i) $r1 100Mb 20ms DropTail
    set n2($i) [$ns node]
    $ns duplex-link $n2($i) $r2 100Mb 20ms DropTail
    set tcp($i) [$ns create-connection TCP/Reno $n1($i) TCPSink $n2($i) $i]
    set ftp($i) [$tcp($i) attach-source FTP]
```

```
# TCP:
set windowVsTimeOne [open WindowVsTimeRenoOne w]
set windowVsTimeAll [open WindowVsTimeRenoAll w]

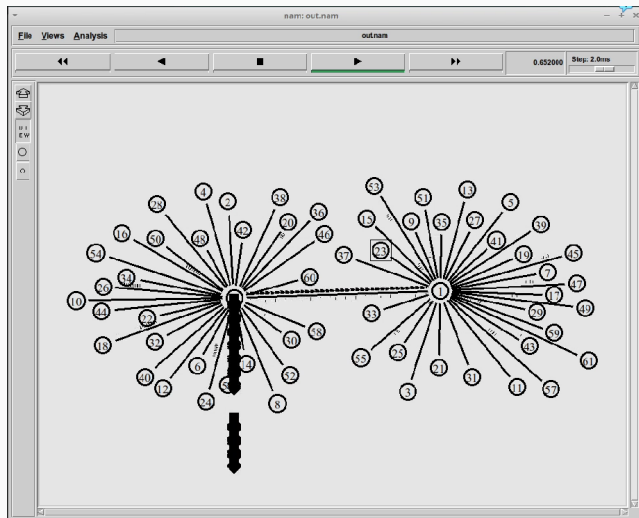
set qmon [$ns monitor-queue $r1 $r2 [open qm.out w] 0.1];
[$ns link $r1 $r2] queue-sample-timeout;

# :
set redq [[$ns link $r1 $r2] queue]
$redq set thresh_ 75
$redq set maxthresh_ 150
$redq set q_weight_ 0.002
$redq set linterm_ 10
```

```
for {set i 0} {$i < $N} {incr i} {  
    $ns at 0.0 "$ftp($i) start"  
    $ns at 0.0 "plotWindow $tcp($i) $windowVsTimeAll"  
}
```

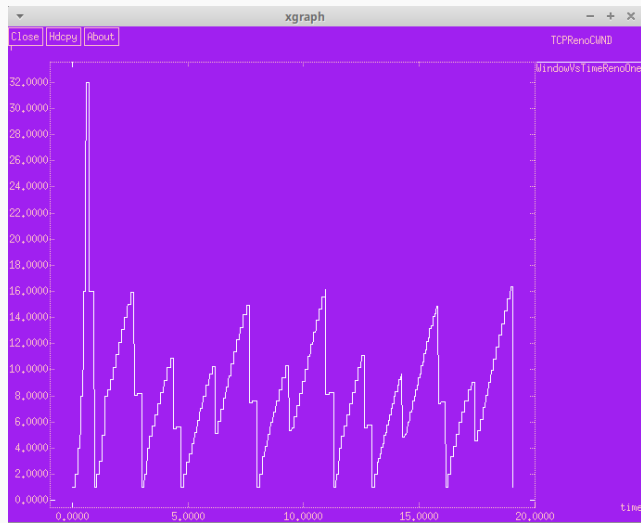
```
$ns at 0.0 "plotWindow $tcp(1) $windowVsTimeOne"
```

```
# at-  
# finish 20s  
$ns at 20.0 "finish"  
#  
$ns run
```



. 1:

N=30

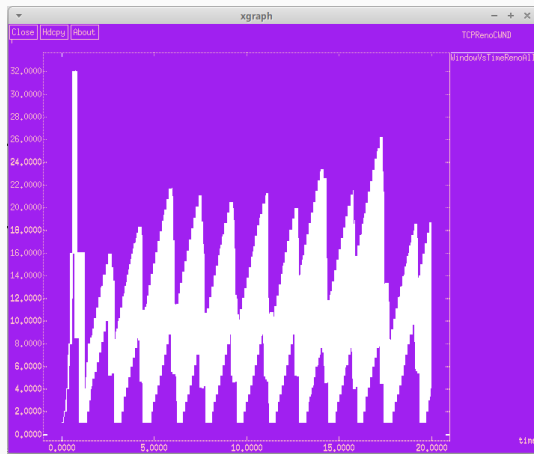


. 2:

TCP

1-

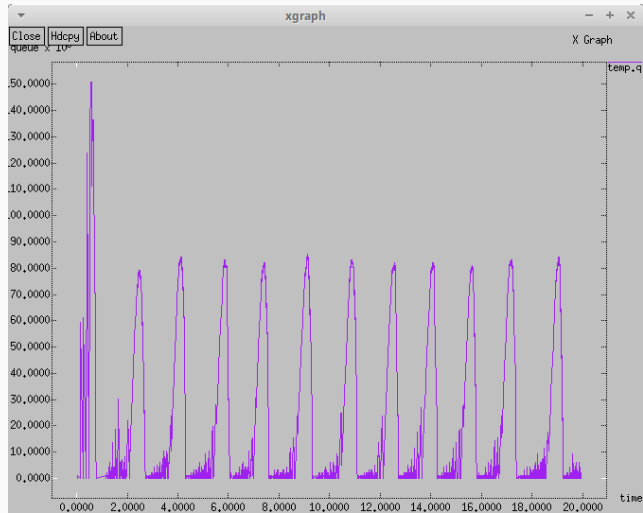
N=30



. 3:

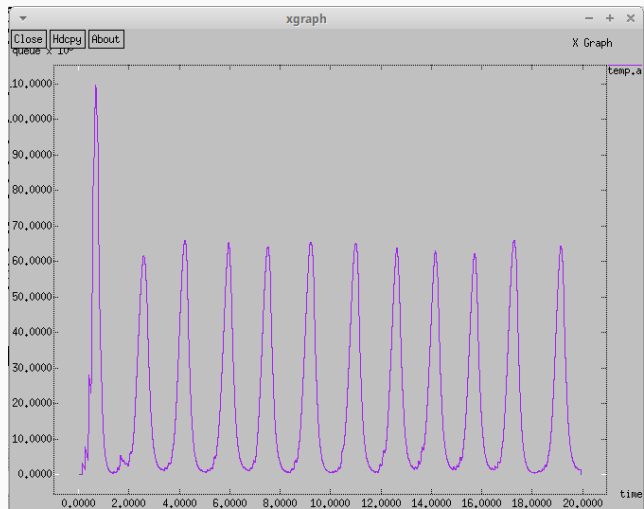
TCP

N=30



. 4:

(R1-R2) N=30



. 5:

(R1-R2) N=30

```
#!/usr/bin/gnuplot -persist
```

```
# ,
```

```
# ,
```

```
set encoding utf8
```

```
set term pngcairo font "Helvetica,9"
```

```
#
set out 'window_1.png'

#
set title "          TCP          1-          N=30"

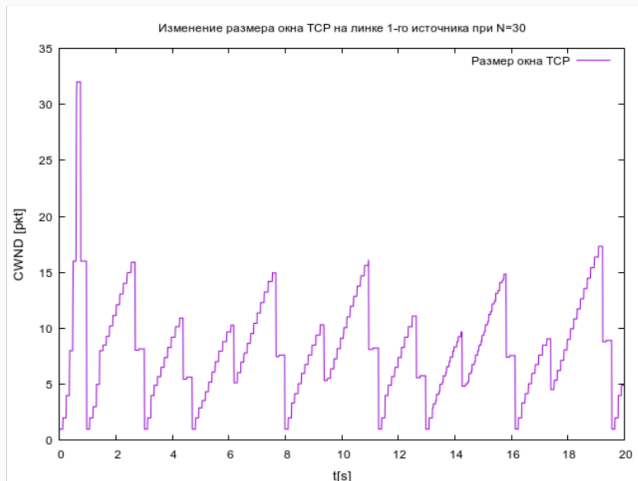
#
set xlabel "t[s]" font "Helvetica, 10"
set ylabel "CWND [pkt]" font "Helvetica, 10"

#
# 1-      2-          WindowVsTimeRenoOne
plot "WindowVsTimeRenoOne" using ($1):($2) with lines title "
```

```
#  
set out 'window_2.png'  
  
#  
set title "          TCP          N          N=30"  
  
#          ,  
# 1-      2-          WindowVsTimeRenoAll  
plot "WindowVsTimeRenoAll" using ($1):($2) with lines title "          TCP"
```

```
#  
set out 'queue.png'  
  
#  
set title "                (R1-R2)"  
  
#  
set xlabel "t[s]" font "Helvetica, 10"  
set ylabel "Queue Length [pkt]" font "Helvetica, 10"  
  
#                ,  
# 1-      2-                temp.q  
plot "temp.q" using ($1):($2) with lines title "
```

```
#  
set out 'av_queue.png'  
  
#  
set title "                                (R1-R2)"  
  
#  
set xlabel "t[s]" font "Helvetica, 10"  
set ylabel "Queue Avg Length [pkt]" font "Helvetica, 10"  
  
#                                ,  
# 1-      2-                temp.a  
plot "temp.a" using ($1):($2) with lines title "
```

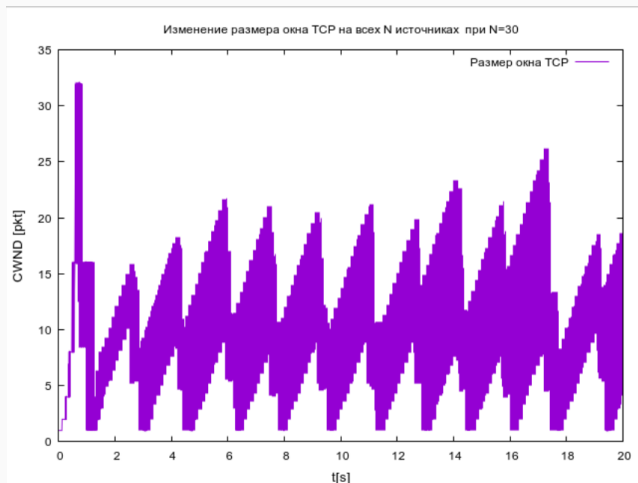



. 6:

TCP

1-

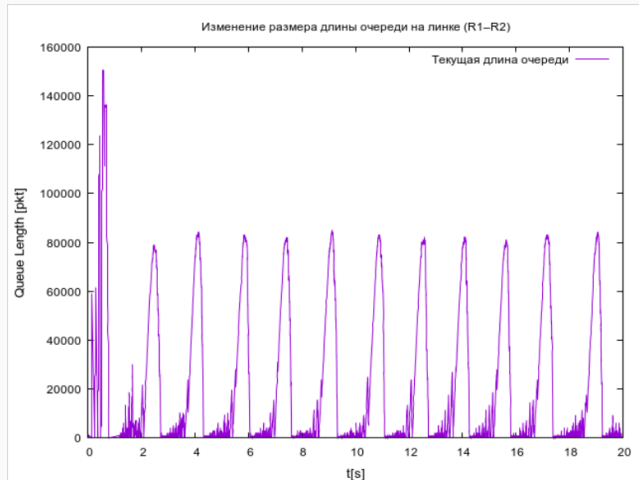
N=30



. 7:

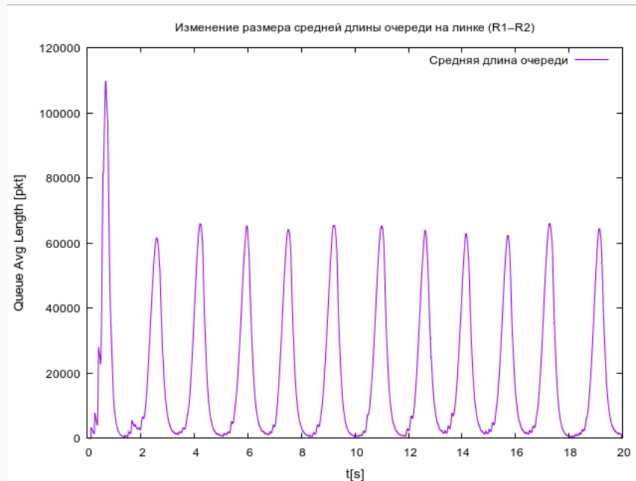
TCP

N=30



. 8:

(R1-R2) N=30



. 9:

(R1-R2) N=30

TCP,

.

NS-2,